

Simple Calculator Feedback

Mistakes & Fixes:

1. Incorrect function calls for multiplication and division

- In these lines:

```
elif msg21 == '*' or msg21 == 'multiplication' or msg21=='multi':  
    print(f"\nYour result is: {subtract(msg6,msg7)}") # Wrong function!
```

Mistake: You're calling subtract() instead of multi().

Fix:

```
print(f"\nYour result is: {multi(msg6,msg7)}")
```

- Similarly, for division:

```
elif msg21 == '/' or msg21 == 'division' or msg21=='multi':  
    print(f"\nYour result is: {subtract(msg6,msg7)}") # Wrong function!
```

Mistake: You're calling subtract() instead of div(), and 'multi' is wrongly included in the division check.

Fix:

```
elif msg21 == '/' or msg21 == 'division' or msg21=='div':  
    print(f"\nYour result is: {div(msg6,msg7)}")
```

2. Loop condition issues (while msg21:)

- while msg21: is used multiple times, but msg21 is a string and will always evaluate as True.
- Fix: Change while msg21: to while True: and handle breaking manually.

3. Division by zero error

- You're allowing division but not checking if the second number is zero, which will cause a crash.
- Fix: Add a check before calling div().

```
elif msg21 == '/' or msg21 == 'division' or msg21=='div':  
    if int(msg7) == 0:  
        print("\nCannot divide by zero. Try again.")  
        continue # Go back to ask for numbers again  
    print(f"\nYour result is: {div(msg6,msg7)}")
```

4. Better input validation for numbers

- isdigit() only works for positive integers and doesn't handle negative numbers or decimals.

- Fix: Use try-except to check for valid numbers instead of isdigit().

try:

```
msg6 = float(input("\nFirst number: "))
```

```
break # Exit loop if input is valid
```

except ValueError:

```
print("\nInvalid input. Enter a number.")
```

Improvements & Suggestions:

- Make code more readable
 - Instead of repeating multiple or conditions, use a dictionary to map operators to functions.

```
operations = {  
    "+": add, "add": add, "addition": add,  
    "-": subtract, "sub": subtract, "subtraction": subtract,  
    "*": multi, "multi": multi, "multiplication": multi,  
    "/": div, "div": div, "division": div  
}
```

- Then replace the long if-elif chain with:

if msg21 in operations:

```
if msg21 in ["/", "division", "div"] and int(msg7) == 0:
```

```
    print("\nCannot divide by zero.")
```

else:

```
    print(f"\nYour result is: {operations[msg21](msg6, msg7)}")
```

else:

```
    print("\nInvalid operator.")
```

- Use while True loops more effectively
 - Instead of using while msg21:, just use while True: and break when needed.
- Fix typos
 - "seconnd number:" should be "second number:"
 - "What numbers woud like to" should be "What numbers would you like to"

Final Thoughts:

Your calculator works, but improving input validation, fixing logic errors, and making the code cleaner will make it more robust and readable. Try applying these fixes and let me know if you need more hints!